

Question	Marking Guidance	Mark	Comments
9.1	1. y axis 0 – 100 in linear scale and x axis minimum 1 to 8 in linear scale and both axes use at least half size of grid; 2. Correct plots for 50% and 25% for both animals; 3. Both curves levelling off (at higher partial pressures and at percentage saturations $\leq 100\%$);	3	1. If tick marks are used on the axis, they must be accurate to within $\pm$ half a small square. 2. 25% - 1.9, 3.3 <u>and</u> 50% - 3.2 and 6.5 2. Accept plot $\pm$ half a small square.
9.2	Correct answer of 15 (times faster) = 2marks ;; If $\geq 3sf$ given, accept answers in the range 15.0 to 15.4 (times faster) = 2marks;; Incorrect answer 1 mark for evidence of: $23^{-0.27}$ divided by $550\,000^{-0.27}$ OR 0.42888777 OR 0.02819045 OR Between 27 and 27.1 OR Between 1.77599861 and 1.8 OR 0.06•	2	Accept any number of significant figures ≥ 2, if rounding correct.

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9.3	1. Mouse <u>haemoglobin/Hb</u> has a lower affinity for oxygen OR For the same pO_2 the mouse <u>haemoglobin/Hb</u> is less saturated OR At oxygen concentrations found in tissue mouse <u>haemoglobin/Hb</u> is less saturated; 2. More oxygen can be dissociated/released/unloaded (for metabolic reactions/respiration);	2	1. For 'Hb is less saturated' accept 'less oxygen will be bound to Hb'. 2. Accept 'oxygen dissociated/released/unloaded more readily/easily/quickly' 2. Reject 'oxygen loaded more readily/easily/quickly' or 'more oxygen loaded'
9.4	Mouse 1. (Smaller so) larger surface area to volume ratio; 2. More/faster heat loss (per gram/in relation to body size); 3. (Faster rate of) respiration/metabolism releases heat;	3	Accept converse answers in relation to the horse. 1. Accept larger SA:V. 1. and 2. must be comparative. 2. Ignore heat lost more easily/readily. 3. Accept respiration/metabolism replaces heat. 3. Reject produce/generate heat/energy.

